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Standards-Compliant Optical Fiber Link Design: From Communication Requirement to Deployment Decision

INTRODUCTION

Optical fiber communication is widely used for high-speed and long-distance data transmission because of its high bandwidth, low attenuation, and resistance to electromagnetic interference.

This engineering design focuses on developing a 60 km optical fiber communication link operating at 1550 nm with a transmission speed of 10 Gb/s and provision for upgrading to 100 Gb/s.

The design involves selecting a suitable optical fiber, calculating numerical aperture and normalized frequency, evaluating optical power loss, analyzing dispersion and polarization effects, and verifying optical signal-to-noise ratio (OSNR).

The final objective is to develop a reliable optical communication system that satisfies the specified transmission requirements.

STAGE 1 – INTERPRET THE COMMUNICATION REQUIREMENT

1.1 Communication Requirements

Parameter	**Requirement**
Transmission distance	60 km
Operating wavelength	1550 nm
Data transmission rate	10 Gb/s
Future upgrade	100 Gb/s
Transmitter power	+3 dBm

Receiver sensitivity	−20 dBm
Required system margin	6 dB
Optical fiber type	ITU-T standard fiber
Building-entry condition	Tight bends
Required OSNR	17 dB

1.2 Importance of 1550 nm Wavelength

The wavelength of 1550 nm is selected because:

1. It provides low transmission loss in silica optical fibers.
2. It is suitable for long-distance optical communication.
3. It supports high-speed data transmission.
4. It is compatible with optical amplifiers such as EDFAs.
5. It supports wavelength-division multiplexing for future capacity upgrades.

1.3 Nature of Light Propagation

Light propagation in optical fiber can be explained using ray theory and wave theory.

Ray Theory:

- Light travels through the fiber core by total internal reflection.
- It is useful for understanding numerical aperture and acceptance angle.
- It provides an approximate explanation of multimode propagation.

Wave Theory:

- Light propagates as guided electromagnetic modes.
- Maxwell's equations describe electromagnetic wave propagation.
- Wave theory is required to analyze LP modes and cutoff conditions.
- It provides an accurate explanation of single-mode propagation.

Coherence and polarization are also important because they influence dispersion, interference, and signal quality.

1.4 Fiber Categories

Fiber Category Characteristics

G.652.D	Standard single-mode optical fiber
G.655	Non-zero dispersion-shifted single-mode fiber
G.657	Bend-insensitive single-mode optical fiber

G.652.D provides standard long-distance transmission.

G.655 offers reduced chromatic dispersion near 1550 nm.

G.657 provides better bending performance, making it suitable for building-entry sections.

1.5 Loss Mechanisms

The important optical losses are:

1. Absorption loss
2. Rayleigh scattering loss
3. Bending loss
4. Connector loss
5. Splice loss

These losses reduce received optical power and must be considered when designing the power budget.

1.6 Dispersion Mechanisms

Modal Dispersion: Different guided modes travel with different delays. It is important in multimode fibers.

Chromatic Dispersion: Different wavelengths travel at different group velocities, causing pulse broadening.

Polarization Mode Dispersion: Different polarization components experience slightly different propagation delays.

Dispersion compensation and suitable fiber selection help maintain pulse integrity.

DESIGN REQUIREMENT STATEMENT

The proposed optical communication system must transmit 10 Gb/s data over 60 km at 1550 nm using standard single-mode optical fiber. It must maintain adequate received optical power, control attenuation and dispersion, satisfy the required OSNR and system margin, tolerate tight building-entry bends, and provide a practical upgrade path to 100 Gb/s.

STAGE 2 – INVESTIGATE AND SELECT THE FIBER CONFIGURATION

2.1 Numerical Aperture

Numerical aperture represents the light-acceptance capability of an optical fiber.

Formula:

$$NA = \sqrt{n_1^2 - n_2^2}$$

Where:

- (n1) = Core refractive index
- (n2) = Cladding refractive index **Fiber A**

Given:

$$n_1 = 1.4500$$

$$n_2 = 1.4450$$

$$NA = \sqrt{1.4500^2 - 1.4450^2}$$

$$NA = \sqrt{2.1025 - 2.088025}$$

$$\boxed{NA_A = 0.1203}$$

Fiber B

Given:

$$n_1 = 1.480$$

$$n_2 = 1.465$$

$$NA = \sqrt{1.480^2 - 1.465^2}$$

$$NA = \sqrt{2.1904 - 2.146225}$$

$$\boxed{NA_B = 0.2102}$$

Comparison

Parameter	Fiber A	Fiber B
Core refractive index	1.450	1.480
Cladding refractive index	1.445	1.465
Core radius	4 µm	25 µm
Numerical aperture	0.1203	0.2102

Fiber B has greater light-acceptance capability. However, a larger numerical aperture does not necessarily make it better for high-speed, long-distance communication.

2.2 V-Number Calculation

The normalized frequency determines the number of guided modes supported by an optical fiber.

Formula:

$$V = \frac{2\pi a}{\lambda} NA$$

Fiber A

$$V_A = \frac{2\pi(4)}{1.55}(0.1203)$$

$$\boxed{V_A = 1.95}$$

Fiber B

$$V_B = \frac{2\pi(25)}{1.55}(0.2102)$$

$$\boxed{V_B = 21.30}$$

2.3 Single-Mode Cutoff Condition

For a step-index optical fiber:

$$V < 2.405$$

Since:

V_A=1.95<2.405

Fiber A supports single-mode propagation.

Since:

V_B=21.30>2.405

Fiber B supports multimode propagation.

2.3 Number of Guided Modes

For a step-index multimode fiber:

$$M \approx \frac{V^2}{2}$$

For Fiber B:

$$M = \frac{21.30^2}{2}$$

$$\boxed{M \approx 227 \text{ modes}}$$

For an approximately parabolic graded-index multimode fiber:

$$M \approx \frac{V^2}{4}$$

$$M \approx 113$$

These are approximate mode counts.

2.4 Cut-Off Wavelength

The single-mode cut-off condition is:

$$V_c = 2.405$$

The cut-off wavelength is:

$$\lambda_c = \frac{2\pi aNA}{2.405}$$

For Fiber A:

$$\lambda_c = \frac{2\pi(4)(0.1203)}{2.405}$$

$$\boxed{\lambda_c \approx 1.257\ \mu m}$$

$$\boxed{\lambda_c \approx 1257\ nm}$$

Since the operating wavelength is 1550 nm, which is greater than the calculated cut-off wavelength, Fiber A supports single-mode operation.

2.5 Comparison of Optical Fiber Types

Fiber type	Characteristics	Application
Single-mode step-index	One spatial propagation mode, no intermodal dispersion	Long-distance communication
Multimode step-index	Many modes, high modal dispersion	Short-distance links
Multimode graded-index	Reduced modal dispersion	Medium-distance communication
G.652.D	Standard single-mode fiber	Long-distance networks
G.655	Non-zero dispersion-shifted single-mode fiber	Long-haul and WDM systems
G.657	Bend-insensitive single-mode fiber	Building entry and tight bends

Ray theory is useful for explaining multimode propagation, while wave theory and LP-mode analysis are necessary to classify guided modes accurately, particularly for Fiber A.

DESIGN DECISION 1

Recommended Configuration: Single-Mode Optical Fiber

I recommend single-mode propagation using a G.655-type fiber for the main 60 km transmission link, with a compatible G.657.A bend-insensitive section at the building entry.

This configuration is recommended because:

- Fiber A demonstrates single-mode operation with \(V=1.95\).
- G.655 provides lower chromatic dispersion at 1550 nm than the specified G.652.D alternative.
- G.657.A provides improved bending performance at the building entry.
- The configuration supports long-distance, high-speed communication.

The calculated properties of Fiber A establish modal suitability; actual purchased fiber must separately be verified as compliant with its specified ITU-T category.

STAGE 3 – DEVELOP AND EVALUATE THE OPTICAL POWER BUDGET

3.1 Total Fiber Attenuation

Optical attenuation is the reduction in optical signal power during transmission through an optical fiber.

The major losses are:

1. Absorption loss
2. Rayleigh scattering loss
3. Bending loss
4. Connector loss
5. Splice loss

3.2 Total Attenuation Coefficient

Given:

Absorption loss:

$$\alpha_a = 0.04\ dB/km$$

Rayleigh scattering loss:

$$\alpha_s = 0.11\ dB/km$$

Distributed bending loss:

$$\alpha_b = 0.03\ dB/km$$

Total attenuation:

$$\alpha = \alpha_a + \alpha_s + \alpha_b$$

$$\alpha = 0.04 + 0.11 + 0.03$$

$$\boxed{\alpha = 0.18\ dB/km}$$

3.4 Fiber Transmission Loss

Formula:

$$L_f = \alpha L$$

$$L_f = 0.18 \times 60$$

$$\boxed{L_f = 10.8\ dB}$$

3.4 Connector Loss

Number of connectors = 4

Loss per connector = 0.4 dB

$$L_c = 4 \times 0.4$$

$$\boxed{L_c = 1.6\ dB}$$

3.5 Splice Loss

Number of splices = 10

Loss per splice = 0.08 dB

$$L_s = 10 \times 0.08$$

$$\boxed{L_s = 0.8\ dB}$$

3.6 Building-Entry Bending Loss

For G.652.D

$$L_b = 0.8\ dB$$

For G.657.A:

$$L_b = 0.1\ dB \quad \mathbf{3.7}$$

Total Link Loss

Formula:

$$L_{total} = L_f + L_c + L_s + L_b$$

Using G.652.D-Type Building Entry

$$L_{total} = 10.8 + 1.6 + 0.8 + 0.8$$

Using G.657.A-Type Building Entry

$$L_{total} = 10.8 + 1.6 + 0.8 + 0.1$$

$$\boxed{L_{total} = 13.3\ dB}$$

3.8 Received Optical Power

Formula:

$$P_r = P_t - L_{total}$$

Transmitter power:

$$P_t = +3\ dBm$$

G.652.D Building Entry

$$P_r = 3 - 14$$

$$\boxed{P_r = -11.0\ dBm}$$

G.657.A Building Entry

$$P_r = 3 - 13.3$$

$$\boxed{P_r = -10.3\ dBm}$$

3.9 System Margin

Formula:

$$M = P_r - P_{sensitivity}$$

Receiver sensitivity:

$$P_{sensitivity} = -20\ dBm$$

For G.652.D entry:

$$M = -11 - (-20)$$

$$\boxed{M = 9\ dB}$$

For G.657.A entry:

$$M = -10.3 - (-20)$$

$$\boxed{M = 9.7\ dB}$$

The required margin is 6 dB.

Therefore, both configurations satisfy the initial optical power requirement.

The G.657.A entry provides an additional 0.7 dB margin.

3.10 Dominant Loss Mechanism

Loss mechanism	Loss over 60 km
Absorption	2.4 dB
Rayleigh scattering	6.6 dB
Distributed bending	1.8 dB
Connectors	1.6 dB
Splices	0.8 dB

Rayleigh scattering is the dominant specified distributed fiber-loss mechanism.

Practical loss-reduction methods include using low-loss fiber and minimizing avoidable connector, splice, and bending losses.

DESIGN DECISION 2

I recommend using a low-loss single-mode optical fiber with G.657.A bend-insensitive fiber at the building entry.

The proposed configuration provides:

- Total link loss: 13.3 dB
- Received optical power: −10.3 dBm
- Available system margin: 9.7 dB

- Required system margin: 6 dB

The initial power budget is suitable for deployment before considering any additional dispersion-compensating components.

STAGE 4 – DISPERSION AND POLARIZATION MANAGEMENT

4.1 Types of Dispersion

Modal Dispersion

Modal dispersion occurs when different propagation modes travel through an optical fiber at different velocities.

It is significant in multimode fibers.

Single-mode fibers eliminate intermodal dispersion.

Chromatic Dispersion

Chromatic dispersion occurs because different wavelength components of an optical signal travel at different group velocities.

It causes optical pulses to broaden during transmission.

Polarization Mode Dispersion

Polarization mode dispersion occurs when two orthogonal polarization components propagate at slightly different speeds.

It can affect high-speed optical communication systems.

4.2 Chromatic Dispersion Calculation

Formula:

$$\Delta\tau = D\Delta\lambda L$$

Given:

$$\Delta\lambda = 0.4\ nm$$

$$L = 60\ km$$

G.652.D Fiber

$$D = 17\ ps/(nm \cdot km)$$

$$\Delta\tau = 17 \times 0.4 \times 60$$

$$\boxed{\Delta\tau = 408\ ps}$$

G.655 Fiber

$$D = 4.5\ ps/(nm \cdot km)$$

$$\Delta\tau = 4.5 \times 0.4 \times 60$$

$$\boxed{\Delta\tau = 108\ ps}$$

4.3 Output Pulse Width

Given input pulse width:

$$\tau_i = 30\ ps$$

Using the root-sum-square approximation:

$$\tau_o = \sqrt{\tau_i^2 + \Delta\tau^2}$$

G.652.D

$$\tau_o = \sqrt{30^2 + 408^2}$$

$$\boxed{\tau_o \approx 409.1\ ps}$$

G.655

$$\tau_o = \sqrt{30^2 + 108^2}$$

$$\boxed{\tau_o \approx 112.1\ ps}$$

Comparison

Parameter	G.652.D	G.655
Dispersion coefficient	17 ps/(nm·km)	4.5 ps/(nm·km)
Pulse broadening	408 ps	108 ps
Output pulse width	409.1 ps	112.1 ps
Bit period	100 ps	100 ps

Both uncompensated pulse widths exceed the 100 ps bit period.

Therefore, neither uncompensated configuration satisfies the pulse-integrity requirement under the assignment's assumed source conditions.

G.655 has significantly lower chromatic dispersion and is the better starting point.

4.4 Dispersion-Compensating Fiber

Given**:**

$$D_{DCF} = -100\ ps/(nm \cdot km)$$

For complete first-order compensation:

$$D_f L_f + D_{DCF} L_{DCF} = 0$$

G.652.D Compensation

$$17(60) - 100L_{DCF} = 0$$

$$L_{DCF} = \frac{1020}{100}$$

$$\boxed{L_{DCF} = 10.2\ km}$$

Alternative: G.655 Compensation

For G.655:

$$4.5(60) - 100L_{DCF} = 0$$

$$\boxed{L_{DCF} = 2.7\ km}$$

4.5 Polarization Mode Dispersion

Formula:

$$PMD = D_{PMD}\sqrt{L}$$

Given:

$$D_{PMD} = 0.1\ ps/\sqrt{km}$$

$$L = 60\ km$$

$$PMD = 0.1\sqrt{60}$$

$$\boxed{PMD = 0.775\ ps}$$

For 10 Gb/s

Bit period:

$$T_b = 100\ ps$$

Maximum allowed PMD:

$$PMD_{limit} = 0.1T_b$$

$$\boxed{PMD}_{limit} \boxed{= 10\ ps}$$

For 100 Gb/s

Using the assignment's approximate bit period:

$$T_b = 10\ ps$$

$$PMD_{limit} = 0.1(10)$$

$$\boxed{PMD}_{limit} \boxed{= 1\ ps}$$

4.6 Birefringence and Beat Length

Birefringence is the difference in effective refractive indices experienced by orthogonal polarization components.

Formula:

$$L_B = \frac{\lambda}{B}$$

The PDF's birefringence exponents are not visible in its extracted text. For the following illustrative calculations, assume:

Standard fiber: $B = 1 \times 10^{-7}$

PM fiber: $B = 4 \times 10^{-4}$

These assumed values should be checked against the original PDF before submission.

Standard Fiber

$$L_B = \frac{1.55 \times 10^{-6}}{1 \times 10^{-7}}$$

$$\boxed{L_B = 15.5\ m}$$

Polarization-Maintaining Fiber

$$L_B = \frac{1.55 \times 10^{-6}}{4 \times 10^{-4}}$$

$$\boxed{L_B = 3.875\ mm}$$

Polarization-maintaining fiber preserves defined polarization axes but is not generally required throughout an ordinary intensity-modulated optical communication link.

DESIGN DECISION 3

I recommend using G.655 single-mode optical fiber with a nominal 2.7 km dispersioncompensating fiber section.

This configuration is preferred because:

- G.655 produces lower chromatic dispersion than G.652.D.
- Only 2.7 km of DCF is required for nominal first-order compensation.
- The additional DCF loss is 1.35 dB.
- The received power is −11.65 dBm.
- The available power margin is 8.35 dB.
- The calculated PMD is within the specified limits.

STAGE 5 – EVALUATE OSNR AND AMPLIFICATION STRATEGY

5.1 Optical Signal-to-Noise Ratio

Optical signal-to-noise ratio measures the ratio of optical signal power to noise power.

Formula:

$$OSNR = P_{signal} - P_{noise}$$

Both powers are expressed in dBm and measured using the same reference bandwidth.

Given:

Signal power:

$$P_s = -16\ dBm$$

ASE noise power:

$$P_n = -39\ dBm$$

Required OSNR:

$$OSNR_{required} = 17\ dB$$

Reference bandwidth:

$$0.1\ nm$$

5.2 Initial OSNR

$$OSNR = -16 - (-39)$$

$$\boxed{OSNR_{initial} = 23\ dB}$$

The initial OSNR exceeds the required value by 6 dB.

5.3 Optical Amplifier Performance

Given:

Signal gain = 14 dB

ASE noise increase = 16 dB

Output Signal Power

$P_{s,out} = -16 + 14$

$$\boxed{P_{s,out} = -2\ dBm}$$

Output Noise Power

$$P_{n,out} = -39 + 16$$

$$\boxed{P_{n,out} = -23\ dBm}$$

5.4 Output OSNR

$$OSNR_{out} = -2 - (-23)$$

$$\boxed{P_{n,out} = -21\ dBm}$$

Comparison

Parameter	Before amplifier	After amplifier
Signal power	−16 dBm	−2 dBm
ASE noise power	−39 dBm	−23 dBm
OSNR	23 dB	21 dB
Required OSNR	17 dB	17 dB
OSNR margin	6 dB	4 dB

The amplifier increases signal power but decreases OSNR by 2 dB.

However, the resulting OSNR remains above the minimum requirement.

5.5 Effect of OSNR on Communication Quality

A low OSNR can cause:

- Increased signal distortion
- Reduced Q-factor
- Increased bit error rate
- Reduced communication reliability
- Reduced system performance margin

Practical OSNR-improvement techniques include:

1. Using optical amplifiers with low noise figures.
2. Reducing unnecessary optical amplification and passive component losses.

DESIGN DECISION 4

I recommend accepting the specified amplifier as an optional amplification stage, subject to a complete system-level OSNR and receiver-power check.

The calculated output OSNR is 21 dB, which exceeds the required 17 dB.

However, the amplifier reduces OSNR by 2 dB. Since the selected passive G.655/DCF design already satisfies the receiver power margin, amplification is not essential for the initial 10 Gb/s power budget.

STAGE 6 – VERIFY AND EVALUATE THE COMPLETE LINK

Requirement	**Target**	**Achieved**	**Status**
Transmission distance	60 km	60 km	Satisfactory

Operating wavelength	1550 nm	1550 nm	Satisfactory
Data rate	10 Gb/s	Designed for 10 Gb/s	Subject to testing
Modal condition	Single-mode	Fiber A: V = 1.95	Satisfactory
Fiber category	Standard ITU-T fiber	G.655 + G.657.A entry	Requires product verification
Fiber attenuation	Low loss	0.18 dB/km	Satisfactory
Total link loss	Within power budget	14.65 dB	Satisfactory
Received power	≥ −20 dBm	−11.65 dBm	Satisfactory
System margin	≥ 6 dB	8.35 dB	Satisfactory
Chromatic dispersion	Controlled	Nominally compensated	Satisfactory in first-order model
DCF length	As required	2.7 km	Satisfactory
PMD at 10 Gb/s	≤ 10 ps	0.775 ps	Satisfactory
PMD at 100 Gb/s	≤ 1 ps	0.775 ps	Satisfactory in simplified model
OSNR	≥ 17 dB	21 dB after amplifier	Satisfactory

Verification Result:

The calculated link meets the main power-budget, first-order dispersion, PMD, and OSNR requirements for the proposed 10 Gb/s service.

The calculations support technical feasibility, but they do not replace practical system testing or demonstrate full 100 Gb/s readiness.

FINAL SYSTEM DESIGN

7.1 Proposed Optical Fiber Communication System

The proposed system uses a single-mode optical fiber link to transmit data between a central office and a business building over 60 km.

A 1550 nm optical transmitter converts electrical data into optical signals.

The optical signals travel through G.655 single-mode optical fiber.

A dispersion-compensating fiber section reduces accumulated chromatic dispersion.

A compatible G.657.A fiber section is used at the building entry to minimize bending loss.

The optical receiver converts the received light into electrical data.

An optical amplifier may be included if required after evaluating the complete receiver-power and OSNR requirements.

7.3 Final Design Parameters

Parameter	**Selected value**
Transmission distance	60 km
Operating wavelength	1550 nm
Data transmission rate	10 Gb/s
Future upgrade target	100 Gb/s
Main fiber category	G.655
Propagation type	Single-mode
Building-entry fiber	G.657.A
Transmitter power	+3 dBm
Fiber attenuation	0.18 dB/km
Nominal DCF length	2.7 km
DCF attenuation	1.35 dB

Total passive link loss	14.65 dB
Received optical power	−11.65 dBm
Parameter	**Selected value**
Receiver sensitivity	−20 dBm
Available power margin	8.35 dB
PMD	0.775 ps
Initial OSNR	23 dB
OSNR after specified amplifier	21 dB
Amplifier gain, if used	14 dB

FINAL ENGINEERING DECISION

Would You Deploy the Proposed Design?

Decision: YES — Conditionally, After Final System Verification

The proposed 60 km optical communication link is technically feasible for 10 Gb/s transmission based on the calculated engineering parameters.

Technical Justification

1. Fiber Selection

G.655 single-mode optical fiber is selected because it provides low chromatic dispersion and supports long-distance high-speed communication.

2. Optical Power Budget

The calculated passive link loss is 14.65 dB.

The received optical power is −11.65 dBm, which is higher than the receiver sensitivity of −20 dBm.

The available system margin is 8.35 dB, exceeding the required 6 dB.

3. Dispersion Compensation

A nominal 2.7 km dispersion-compensating fiber section is used to cancel the first-order accumulated chromatic dispersion.

4. Polarization Mode Dispersion

The calculated PMD is approximately 0.775 ps, which is below the specified limits for 10 Gb/s and the simplified 100 Gb/s upgrade case.

5. Signal Quality

The calculated OSNR after the specified amplifier is 21 dB, exceeding the required 17 dB.

6. Building-Entry Bending Loss

G.657.A bend-insensitive fiber reduces the specified building-entry bending loss from 0.8 dB to 0.1 dB.

7. Future Upgrade

The single-mode infrastructure provides a potential path toward 100 Gb/s. However, upgrading would require verification of the transmitter, receiver, modulation format, chromatic-dispersion management, PMD tolerance, and OSNR.

Final Conclusion

The proposed optical fiber communication system uses G.655 single-mode fiber, a G.657.A bend-insensitive building-entry section, and a nominal 2.7 km dispersion-compensating fiber section.

The calculated design provides a passive link loss of 14.65 dB, received optical power of −11.65 dBm, and an available power margin of 8.35 dB.

The calculated PMD and OSNR also satisfy the specified numerical limits.

Therefore, **the proposed system is recommended for conditional deployment at 10 Gb/s over 60 km**, subject to equipment compatibility checks, actual loss and dispersion measurements, OSNR verification, and end-to-end BER testing.

The system provides a potential foundation for a future 100 Gb/s upgrade, but that upgrade must be engineered and verified separately.